

Achilles Tendinopathy

Background

- Achilles tendinopathy is caused by overuse of your Achilles tendon. It can affect both athletes and non-athletes alike.
- Diagnosis is made through a review of your history and an exam that commonly includes an ultrasound evaluation and X-Rays. Damage seen within the tendon confirms the diagnosis.
- Ultrasound imaging is more accurate than MRIs for diagnosing tendon issues because it provides clearer, moving images.
- Intrinsic risk factors can include changed bio-mechanics, calf-muscle weakness, limited ankle mobility, ages 30–60, and obesity.
- Extrinsic risk can include joint or muscle overuse, large increases in running mileage in a short period of time, too little time taken for recovery after exercise, or immediate changes in footwear without gradually changing over.

Latest Research

- Rehab exercises that involve eccentric motion (or "controlled lowering") have proven to be effective for treatment.
- Loading (strain on the Achilles tendon) must be progressed gradually. Don't go for a long run or do heavy jumping exercises too soon.
- Recent studies show that treatment with nitroglycerin patches can be helpful for reducing pain.
- If rehab is done correctly but symptoms persist, there are additional options that might help you. Either dextrose prolotherapy injections—a type of shot that "jumpstarts" your body's healing response—or percutaneous ultrasonic tenotomy, which is a nonsurgical way to remove damaged tendon tissue, might help.

Practical Tips for Success

Load-reduction phase (2–4 weeks)

It's best to reduce the load on your body (cut out all running and jumping activities if you can) and introduce alternative, low-level activities when recovering from Achilles tendinopathy.

Gradual-load phase (2–6 months)

After symptoms improve with load reduction, you can gradually increase the load with eccentric exercises (for example, the single-leg heel lowers described above).

- The most common pitfall (other than giving up!) is increasing the load too quickly. Be patient, be consistent, and progress gradually.
- Heel-lift shoe inserts might be helpful for short-term use (2–4 weeks). But these lifts can cause problems if worn for longer than 4 weeks.
- Tendons take time to heal. Full recovery takes 3–6 months, often longer. Again, be patient, be consistent, and progress gradually.

Rehab Instructions

Calf Stretch:

❶ Use a wall for support, and stand with one foot about half a step behind your other foot.

❷ Lean forward until you feel a stretch in your calf on the back leg.

❸ Hold for 30–60 seconds.

☐ Complete 3 sets on each side 1x a day before you do the exercises below.

Double-leg heel raises:

❶ Start with your feet flat on floor.

❷ Quickly raise up onto the balls of your feet as high as you can.

❸ Then, slowly lower both heels back to the floor.

☐ Complete 3 sets of 15 reps 1x or 2x a day for 4–12 weeks.

- Advance to single-leg heel lowers when double-leg heel raises are easy and painless.

Single-leg heel lowers (eccentrics):

❶ Start with both feet flat on floor.

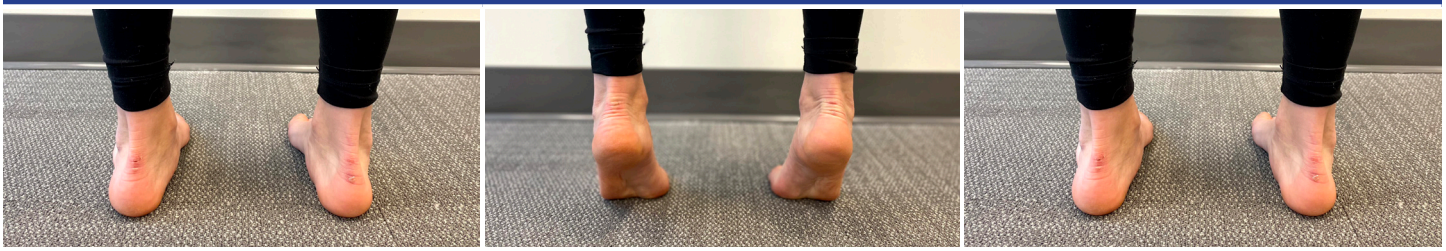
❷ Raise your heels to stand on the balls of your feet.

❸ Then, put all of your weight on the injured foot and slowly lower your heel to the ground.

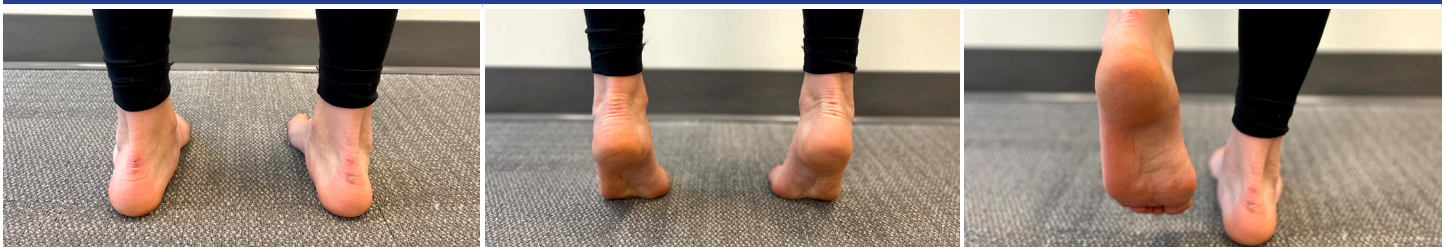
☐ Complete 3 sets of 15 reps 2x a day for 6–12 weeks.

- You can modify the exercise by adding a slight bend to the knee of your injured leg.
- When the exercise becomes easy and not painful, you may gradually add weight.
- To add weight, use a backpack and add books, canned food, full water bottles, or similar items.
- Unlike most rehab exercises, single-leg eccentrics should be slightly painful to get the maximum benefit from the exercise.

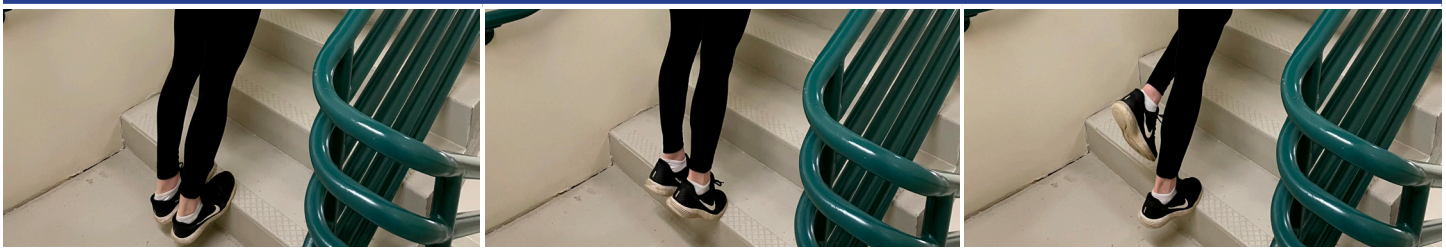
Double-leg heel raises | Both feet on flat surface



Single-leg heel lowers (eccentrics) | Up on both feet, lower down on one foot (the injured side)



Eccentrics progression | Add weight, and/or add step



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